CORE PURE / FURTHER CALCULUS

# Improper Integrals and Mean Values

Mark scheme 16 questions · 116 marks

## Questions

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| | **Total** | **116** | |

# Reading the mark scheme

These rules apply throughout, unless a note says otherwise.

### Marks

Every mark is worth one, so **M1 A1** on a row is two marks. Only marks coded ft follow through.

| Code | Meaning |
|---|---|
| **M1** | Method: a correct method, applied to this question. A slip doesn't cost it if the method is clear. A formula must be used, not just quoted. |
| **dM1** | Dependent method: as M1, but only if the method mark before it was earned, unless its note names another. |
| **A1** | Accuracy: a correct answer or step. It needs the method mark it follows. |
| **A1ft** | As A1, following through from an earlier answer that the note names. |
| **A1 AG** | The last mark of a part whose answer is printed in the question: complete working, with no errors. |
| **B1** | A correct value, fact or statement. It needs no method mark. |
| **B1ft** | As B1, following through from an earlier answer that the note names. |
| **E1** | A written explanation, interpretation or conclusion. Concluding a proof by induction needs every earlier mark. |

### Abbreviations

ft and AG qualify a code in the Marks column. The others appear in the notes.

| | | | |
|---|---|---|---|
| **ft** | follow through | **AG** | answer given in the question |
| **cao** | correct answer only, with no follow-through | **cso** | correct solution only: no errors in the part |
| **oe** | or an equivalent form | **awrt** | any answer that rounds to this value |
| **soi** | seen, or implied by later working | **isw** | ignore working after a correct answer |
| **SC** | special case: credit outside the main method | **their** | the student's own earlier value |
| **condone** | a slip that doesn't cost the mark | [ ] | working the mark doesn't need |
| **s.f.** | significant figures | **d.p.** | decimal places |

### Reading the scheme

- Each row is a line of working, with the marks it earns beside it. Where a row carries two marks, each note starts with its code.
- **[*n*]** ends each part with its total.
- A shaded *Alternative* is a second method for the same total. Mark an answer by the method it follows.
- Any other valid method earns the equivalent marks, unless the question names a method or says *hence*.
- A sketch earns one B1 for each feature named below its figure.

### Accuracy and answers

- Unless a question says otherwise, give answers to 3 significant figures, and angles in degrees to 1 decimal place. Angles in radians take 3 significant figures.
- A final answer with more figures is accepted if it rounds to the scheme's value; one with fewer is not. An answer spoiled by rounding too early loses its accuracy mark.
- Where a question asks for an exact answer, a decimal loses the final accuracy mark. Otherwise either is accepted.
- Wrong or missing units cost nothing unless the question asks for them.
- An omitted $\mathrm{d}\theta$, $\mathrm{d}x$ or $\mathrm{d}u$ in an integral is not penalised where the variable of integration is clear from the integrand and the limits. It is penalised where it is genuinely ambiguous: a substitution part-way through, two variables in play, or limits in the wrong variable.
- A correct answer with no working earns full marks, unless the question needs the working: show that, prove, hence, a method it names, or an exact answer. A wrong answer with no working earns nothing. Calculator answers are treated the same way.
- A correct answer, once reached, isn't spoiled by later working (isw).

### Value of $g$

- Take $g = 9.8\,\mathrm{m\,s^{-2}}$ unless a question states a value, which is then the only one accepted. Answers that use $g = 9.8$ may be given to 2 or 3 significant figures.
- Where a question doesn't state $g$, an answer worked with $g = 10$ throughout earns the same marks, keeping any value printed in the question. The worked solutions list the final answers with $g = 10$.

| Question | Scheme | Marks | Notes |
|---|---|---|---|
| **1** | $\int \frac{x}{x^2+1}\,\mathrm{d}x = \frac{1}{2}\ln(x^2+1)$ | **M1 A1** | **M1** Integrates to the form $k\ln(x^2+1)$, or $k\ln u$ after substituting $u = x^2+1$.<br>**A1** oe, such as $\frac{1}{2}\ln u$.<br>Condone modulus signs. |
| | $\frac{1}{3-1}\left(\text{"}\frac{1}{2}\text{"}\ln 10 - \text{"}\frac{1}{2}\text{"}\ln 2\right)$ | **dM1** | Substitutes limits 1 and 3 (or 2 and 10 in $u$) into their $k\ln(x^2+1)$, subtracts, and divides by 2. |
| | $\frac{1}{4}\ln 5$ | **A1** | Must be in the form $p\ln q$; accept $\frac{1}{8}\ln 25$. A0 for $\frac{1}{4}(\ln 10 - \ln 2)$ or a decimal. |
| | **Total for Question 1** | **4** | |

| Question | Scheme | Marks | Notes |
|---|---|---|---|
| **2** | $\int x(x^2+5)^{-\frac{3}{2}}\,\mathrm{d}x = k(x^2+5)^{-\frac{1}{2}}$ | **M1** | Integrates to the form $k(x^2+5)^{-\frac{1}{2}}$, $k \neq 0$, by reverse chain rule or substitution $u = x^2+5$. |
| | $= -(x^2+5)^{-\frac{1}{2}}\ (+c)$ | **A1** | oe, such as $-\frac{1}{\sqrt{u}}$ with $u = x^2+5$. |
| | $\int_2^t \ldots \mathrm{d}x = \left[-\frac{1}{\sqrt{x^2+5}}\right]_2^t$<br>$= -\frac{1}{\sqrt{t^2+5}} + \frac{1}{3}$ | **M1** | Replaces $\infty$ by $t$ (any letter) and substitutes $t$ and 2, or corresponding limits in $u$, into their $k(x^2+5)^{-\frac{1}{2}}$, subtracting. |
| | As $t \to \infty$, $\frac{1}{\sqrt{t^2+5}} \to 0$ | **dM1** | States that their term in $t$ tends to 0 as $t \to \infty$, in limit notation or words. Not for substituting $\infty$ directly. |
| | $\int_2^\infty \ldots \mathrm{d}x = \frac{1}{3} - 0 = \frac{1}{3}$ | **A1 AG** | cso: correct antiderivative and limits leading without error to $\frac{1}{3}$, with the limiting process shown. |
| | **Total for Question 2** | **5** | |

| Question | Scheme | Marks | Notes |
|---|---|---|---|
| **3** | $\dfrac{4}{x(x+1)(x+2)} = \dfrac{A}{x} + \dfrac{B}{x+1} + \dfrac{C}{x+2}$ | **M1** | Uses the correct form with three constants. Not for a form missing a term. |
| | $4 = A(x+1)(x+2) + Bx(x+2) + Cx(x+1)$<br>$\Rightarrow A = \ldots,\ B = \ldots,\ C = \ldots$ | **dM1** | Finds all three constants, by substituting values or equating coefficients. Condone arithmetic slips. |
| | $\dfrac{2}{x} - \dfrac{4}{x+1} + \dfrac{2}{x+2}$ | **A1** | |
| | $2\ln x - 4\ln(x+1) + 2\ln(x+2)$ | **M1 A1** | **M1** Integrates at least two of their partial fractions of the form $\frac{a}{x+b}$ to $a\ln(x+b)$. Modulus signs not needed.<br>**A1** All three correct; need not be combined. |
| | $\left[2\ln\dfrac{x(x+2)}{(x+1)^2}\right]_1^t$<br>$= 2\ln\dfrac{t(t+2)}{(t+1)^2} - 2\ln\dfrac{3}{4}$ | **M1** | Replaces $\infty$ by $t$ or uses a limit, combines their log terms into one logarithm of a function of $t$, and substitutes both limits. Condone sign slips. |
| | As $t \to \infty$, $\dfrac{t^2+2t}{t^2+2t+1} = \dfrac{1+\frac{2}{t}}{1+\frac{2}{t}+\frac{1}{t^2}} \to 1$<br>$\Rightarrow 2\ln(\ldots) \to 0$, so the integral converges | **dM1** | Shows the argument of the combined logarithm tends to 1 (e.g. dividing by $t^2$), so its log tends to 0, and concludes convergence. Not for 0 written without a limit argument. |
| | $\ln\dfrac{16}{9}$ | **A1** | cso. Must be in the form $\ln k$, or $k = \frac{16}{9}$ stated; $2\ln\frac{4}{3}$ alone does not score. |
| | **Total for Question 3** | **8** | |

| Question | Scheme | Marks | Notes |
|---|---|---|---|
| **4** | $y = \dfrac{1+\sin x}{1-\sin^2 x} = \dfrac{1+\sin x}{\cos^2 x}$ | **M1** | Multiplies numerator and denominator by $1+\sin x$ and uses $1-\sin^2 x = \cos^2 x$ to reach a denominator of $\cos^2 x$. |
| | $y = \sec^2 x + \sec x \tan x$ | **A1** | |
| | $\displaystyle\int \left(\sec^2 x + \sec x \tan x\right) \mathrm{d}x = \tan x + \sec x$ | **M1 A1** | **M1** Integrates their expression of the form $p\sec^2 x + q\sec x \tan x$ to the form $a\tan x + b\sec x$.<br>**A1** A1: Correct $\tan x + \sec x$. |
| | $\dfrac{6}{\pi}\left[\left(\frac{1}{\sqrt{3}} + \frac{2}{\sqrt{3}}\right) - (0+1)\right]$ | **dM1** | Substitutes both limits into their integral, subtracts, and divides by $\frac{\pi}{6}$. Need not be simplified. |
| | $\dfrac{6\left(\sqrt{3}-1\right)}{\pi}$ | **A1** | cso. Any equivalent exact form. |
| | **Total for Question 4** | **6** | |

| Question | Scheme | Marks | Notes |
|---|---|---|---|
| **5** | $\frac{1}{2}\int_0^2 40\tanh^2\left(\frac{1}{2}t\right)\mathrm{d}t$<br>$= 20\int_0^2\left(1-\operatorname{sech}^2\left(\frac{1}{2}t\right)\right)\mathrm{d}t$ | **M1** | Uses a hyperbolic identity, such as $\tanh^2 x = 1-\operatorname{sech}^2 x$, to replace $\tanh^2\left(\frac{1}{2}t\right)$ in the integral of $p$. Condone a sign slip in the identity. |
| | $20\left[t-2\tanh\left(\frac{1}{2}t\right)\right]_0^2$ | **M1 A1** | **M1** Integrates to the form $\alpha t+\beta\tanh\left(\frac{1}{2}t\right)$.<br>**A1** Correct, with factor 20, or 40 if not yet divided by 2. |
| | $20\left(2-2\tanh 1\right)-20(0)\ [=40(1-\tanh 1)]$ | **dM1** | Substitutes both limits into their integral and divides by 2. Lower limit giving 0 need not be shown. |
| | $\tanh 1=\frac{e-e^{-1}}{e+e^{-1}}\ \left[=\frac{e^2-1}{e^2+1}\right]$ | **M1** | Writes $\tanh 1$, or $\sinh 1$ and $\cosh 1$, in exponential form in their expression. Need not be simplified. |
| | $\frac{80}{e^2+1}$ | **A1** | cso. Only this form scores, not e.g. $\frac{80e^{-2}}{1+e^{-2}}$ or $40(1-\tanh 1)$. |
| | **Total for Question 5** | **6** | |

| Question | Scheme | Marks | Notes |
|---|---|---|---|
| **6(a)** | Upper limit is infinite, so the interval of integration is unbounded | **E1** | States that the upper limit is infinite, or the interval unbounded. Not for a reason about the integrand alone. |
| | | **[1]** | |
| **6(b)** | $u = \ln x,\ \frac{\mathrm{d}v}{\mathrm{d}x} = x^{-3} \Rightarrow v = -\frac{1}{2x^2}$<br>$\int \frac{\ln x}{x^3}\,\mathrm{d}x = -\frac{\ln x}{2x^2} + \int \frac{1}{2x^3}\,\mathrm{d}x$ | **M1 A1** | **M1** Integrates by parts with $u = \ln x$, reaching the form $px^{-2}\ln x \pm \int qx^{-3}\,\mathrm{d}x$. Condone sign and coefficient slips.<br>**A1** Correct first stage; need not be simplified. |
| | $= -\frac{\ln x}{2x^2} - \frac{1}{4x^2} + c$ | **A1** | oe. Needs $+c$. |
| | | **[3]** | |
| **6(c)** | $\int_e^t \frac{\ln x}{x^3}\,\mathrm{d}x = \left[\text{"} - \frac{\ln x}{2x^2} - \frac{1}{4x^2}\text{"}\right]_e^t$ | **M1** | Replaces $\infty$ by $t$ and substitutes $t$ and $e$ into their (b) of the form $px^{-2}\ln x + qx^{-2}$. Not for substituting $\infty$ directly. |
| | $= \frac{3}{4e^2} - \frac{\ln t}{2t^2} - \frac{1}{4t^2}$ | **A1** | oe; $\ln e$ need not be evaluated. |
| | As $t \to \infty$, $\frac{\ln t}{t^2} \to 0$ and $\frac{1}{t^2} \to 0$ | **dM1** | Lets $t \to \infty$, stating explicitly that $\frac{\ln t}{t^2} \to 0$ (oe) and $\frac{1}{t^2} \to 0$. Not for replacing the terms by 0 without a statement. |
| | Converges: $\int_e^\infty \frac{\ln x}{x^3}\,\mathrm{d}x = \frac{3}{4e^2}$ | **A1** | cso. Exact value oe, with a statement that the integral converges or the limit is finite. |
| | | **[4]** | |
| | **Total for Question 6** | **8** | |

| Question | Scheme | Marks | Notes |
|---|---|---|---|
| **7(a)** | $9(x^2-2x)+25=9\left[(x-1)^2-1\right]+25$ | **M1** | Takes out 9 from the $x$ terms and completes the square to $9(x\pm1)^2+q$, or compares coefficients of $a(x+p)^2+q$. Condone a slip in $q$. |
| | $=9(x-1)^2+16$ | **A1** | Or $a=9$, $p=-1$, $q=16$ stated. |
| | | **[2]** | |
| **7(b)** | $\int \frac{\mathrm{d}x}{\sqrt{9(x-1)^2+16}}$<br>$=\frac{1}{3}\int \frac{\mathrm{d}x}{\sqrt{(x-1)^2+\left(\frac{4}{3}\right)^2}}$ | **M1** | Uses their (a), of the form $9(x+p)^2+q$ with $q>0$, with the arsinh result to reach $\lambda\,\mathrm{arsinh}(\mu(x+p))$ or a logarithmic equivalent. Condone slips in $\lambda$, $\mu$. |
| | $=\frac{1}{3}\,\mathrm{arsinh}\left(\frac{3(x-1)}{4}\right)+c$ | **A1** | oe, including a logarithmic form. Needs $+c$. |
| | | **[2]** | |
| **7(c)** | $\frac{1}{2}\times\frac{1}{3}\left[\mathrm{arsinh}\left(\frac{3(x-1)}{4}\right)\right]_0^2$<br>$=\frac{1}{6}\left(\mathrm{arsinh}\,\frac{3}{4}-\mathrm{arsinh}\left(-\frac{3}{4}\right)\right)$ | **M1** | Substitutes 0 and 2 into their (b), of the form $\lambda\,\mathrm{arsinh}(\mu(x+p))$ or a logarithmic equivalent, subtracts and divides by 2. Condone sign slips. |
| | $\mathrm{arsinh}\,\frac{3}{4}=\ln\left(\frac{3}{4}+\sqrt{\frac{9}{16}+1}\right)=\ldots$ | **M1** | Uses $\mathrm{arsinh}\,t=\ln\left(t+\sqrt{t^2+1}\right)$ on at least one of their values, or evaluates a logarithmic form from (b) at a limit. |
| | $\frac{1}{6}(\ln 2+\ln 2)=\frac{1}{3}\ln 2$ | **A1** | cso. Any form $A\ln B$ with $B$ an integer, such as $\frac{1}{6}\ln 4$. Not $\frac{1}{6}\left(\ln 2-\ln\frac{1}{2}\right)$. |
| | | **[3]** | |
| | **Total for Question 7** | **7** | |

| Question | Scheme | Marks | Notes |
|---|---|---|---|
| **8(a)** | $\int \ln(1+x)\,\mathrm{d}x = x\ln(1+x) - \int \frac{x}{1+x}\,\mathrm{d}x$ | **M1** | Applies parts with $u = \ln(1+x)$, reaching $px\ln(1+x) - \int \frac{qx}{1+x}\,\mathrm{d}x$, or the equivalent after substituting $t = 1+x$. Condone sign slips. |
| | $= x\ln(1+x) - x + \ln(1+x)$ | **A1** | oe, such as $(1+x)\ln(1+x) - x$ or $t\ln t - t$. |
| | $\frac{1}{e-1}\Big[(1+x)\ln(1+x) - x\Big]_0^{e-1}$<br>$= \frac{1}{e-1}\big((e\ln e - (e-1)) - 1\times\ln 1\big)$ | **dM1** | Substitutes both limits, or 1 and $e$ after substituting $t = 1+x$, into their antiderivative and divides by $e-1$. |
| | $= \frac{1}{e-1}$ | **A1 AG** | Correct substituted expression, using $\ln e = 1$ and $\ln 1 = 0$, leading without error to the printed result. |
| | | **[4]** | |
| **8(b)** | $x = e^y - 1$ | **B1** | |
| | $\frac{1}{1-0}\int_0^1 (e^y - 1)\,\mathrm{d}y = \Big[e^y - y\Big]_0^1 = \ldots$ | **M1** | Integrates their $x$ in terms of $y$, at least one term correct, and substitutes $y = 1$ and $y = 0$. Division by $1-0$ need not be shown. |
| | $e - 2$ | **A1** | |
| | | **[3]** | |
| | **Total for Question 8** | **7** | |

| Question | Scheme | Marks | Notes |
|---|---|---|---|
| **9** | $\frac{n}{2\pi}\int_0^{2\pi/n}\left(a^2\sin^2 nt + 2ab\sin nt\sin 2nt + b^2\sin^2 2nt\right)\mathrm{d}t$ | **M1** | Forms the integral of $I^2$, expanded into three terms, from 0 to $\frac{2\pi}{n}$, with or without $\frac{n}{2\pi}$. Condone a slip in the cross term's coefficient. |
| | $\int\sin^2 nt\,\mathrm{d}t = \int\frac{1}{2}(1-\cos 2nt)\,\mathrm{d}t = \frac{t}{2} - \frac{\sin 2nt}{4n}$ | **M1** | Uses $\sin^2 x = \frac{1}{2}(1\pm\cos 2x)$ for at least one square and integrates to the form $pt + q\sin knt$. Condone sign and coefficient slips. |
| | $\int_0^{2\pi/n}\sin^2 nt\,\mathrm{d}t = \int_0^{2\pi/n}\sin^2 2nt\,\mathrm{d}t = \frac{\pi}{n}$ | **A1** | |
| | $\int\sin nt\sin 2nt\,\mathrm{d}t = \int 2\sin^2 nt\cos nt\,\mathrm{d}t = \frac{2\sin^3 nt}{3n}$ | **M1** | Integrates the cross term by a valid method, e.g. to the form $k\sin^3 nt$, or by product-to-sum to $p\sin nt + q\sin 3nt$. Condone coefficient slips. |
| | $\left[\frac{2\sin^3 nt}{3n}\right]_0^{2\pi/n} = 0$ | **A1** | Limits substituted into a correct integral; a bare statement that the integral is zero does not score. |
| | $\frac{n}{2\pi}\left(a^2\times"\frac{\pi}{n}" + 2ab\times"0" + b^2\times"\frac{\pi}{n}"\right)$ $\Rightarrow \text{RMS} = \sqrt{\ldots}$ | **dM1** | Combines their three integrals with their coefficients, multiplies by $\frac{n}{2\pi}$ and takes the square root. Depends on the first M1. |
| | $\frac{a^2+b^2}{2} \Rightarrow \text{RMS} = \sqrt{\frac{a^2+b^2}{2}}$ | **A1 AG** | cso: correct integrals combined without error to the printed result. |
| | **Total for Question 9** | **7** | |

| Question | Scheme | Marks | Notes |
|---|---|---|---|
| **10** | $\dfrac{x+3}{x^2(x+1)} = \dfrac{A}{x} + \dfrac{B}{x^2} + \dfrac{C}{x+1}$ | **M1** | Uses partial fractions with separate terms in $x$ and $x^2$ for the repeated factor, and a term in $x+1$. |
| | $x+3 = Ax(x+1) + B(x+1) + Cx^2$<br>$\Rightarrow A = \ldots,\ B = \ldots,\ C = \ldots$ | **dM1** | Clears fractions and finds all three constants, by substitution or equating coefficients. Condone arithmetic slips. |
| | $-\dfrac{2}{x} + \dfrac{3}{x^2} + \dfrac{2}{x+1}$ | **A1** | |
| | $\left[-2\ln x - \dfrac{3}{x} + 2\ln(x+1)\right]$ | **M1 A1** | **M1** Integrates their partial fractions to the form $p\ln x + \frac{q}{x} + r\ln(x+1)$. Condone sign and coefficient slips.<br>**A1** Correct antiderivative; modulus signs optional. |
| | $\lim_{t\to\infty}\left[2\ln\left(\dfrac{x+1}{x}\right) - \dfrac{3}{x}\right]_1^t$<br>$= \lim_{t\to\infty}\left(2\ln\left(1+\frac{1}{t}\right) - \frac{3}{t} - 2\ln 2 + 3\right)$ | **M1** | Uses an upper limit $t$, combines their two log terms into one logarithm, before or after substituting, and substitutes both limits. Not if $\infty$ is substituted. |
| | $t\to\infty:\ \ln\left(1+\frac{1}{t}\right)\to\ln 1 = 0,\ \frac{3}{t}\to 0$ | **dM1** | States, as $t\to\infty$, that their combined log term tends to $\ln 1 = 0$ and their $\frac{k}{t}$ term tends to 0. |
| | $\displaystyle\int_1^\infty \frac{x+3}{x^2(x+1)}\,\mathrm{d}x = 3 - 2\ln 2 = 3 - \ln 4$ | **A1 AG** | cso, from correct limits leading without error to $3-\ln 4$. Withhold if $\infty$ is substituted or $\ln\infty$ terms are cancelled anywhere. |
| | **Total for Question 10** | **8** | |

| Question | Scheme | Marks | Notes |
|---|---|---|---|
| **11(a)** | Upper limit is infinite, so the interval of integration is unbounded | **E1** | States that the upper limit is infinite, or the interval is unbounded. Not for a reference to the integrand alone. |
| | | **[1]** | |
| **11(b)** | $\frac{10}{(x+2)(3x+1)} = \frac{A}{x+2} + \frac{B}{3x+1}$<br>$10 = A(3x+1) + B(x+2) \Rightarrow A = \ldots,\ B = \ldots$ | **M1** | Uses the correct partial-fraction form and a method for both constants, by substitution or comparing coefficients. |
| | $A = -2,\ B = 6$ | **A1** | |
| | $\int \left(-\frac{2}{x+2} + \frac{6}{3x+1}\right) \mathrm{d}x$<br>$= p\ln(x+2) + q\ln(3x+1)$ | **M1** | Integrates their two partial fractions to the form $p\ln(x+2) + q\ln(3x+1)$. Condone coefficient slips and missing modulus signs. |
| | $= 2\ln\left(\frac{3x+1}{x+2}\right)$ | **A1** | Correct single logarithm, oe; may first be seen after the limits are substituted. |
| | $\left[2\ln\frac{3x+1}{x+2}\right]_0^t = 2\ln\frac{3t+1}{t+2} - 2\ln\frac{1}{2}$<br>$\frac{3t+1}{t+2} = \frac{3+\frac{1}{t}}{1+\frac{2}{t}} \to \ldots$ as $t \to \infty$ | **dM1** | Replaces $\infty$ by a variable, substitutes both limits and finds the limit of their logarithm's argument as it tends to infinity. Not for substituting $\infty$ directly. |
| | $2\ln 3 + 2\ln 2 = 2\ln 6$ | **A1** | cso. Any correct $a\ln b$ with integer $a$ and rational $b$, or $a = 2$, $b = 6$ stated. |
| | | **[6]** | |
| | **Total for Question 11** | **7** | |

| Question | Scheme | Marks | Notes |
|---|---|---|---|
| **12(a)** | At $x = 1$, $\sqrt[3]{x^2 - 1} = 0$: integrand undefined<br>$x = 1$ lies in $[0, 3]$ | **E1** | States the integrand is undefined or unbounded (or denominator zero) at $x = 1$, and that 1 lies within the interval of integration. |
| | | **[1]** | |
| **12(b)** | $\int x\left(x^2 - 1\right)^{-\frac{1}{3}}\,\mathrm{d}x = k\left(x^2 - 1\right)^{\frac{2}{3}}$ | **M1** | Integrates to the form $k\left(x^2 - 1\right)^{\frac{2}{3}}$, or $ku^{\frac{2}{3}}$ with $u = x^2 - 1$, by inspection or substitution. |
| | $= \frac{3}{4}\left(x^2 - 1\right)^{\frac{2}{3}}\ (+c)$ | **A1** | oe, including $\frac{3}{4}u^{\frac{2}{3}}$ with $u = x^2 - 1$. |
| | Splits at $x = 1$:<br>$\lim_{t\to 1^-}\left[\frac{3}{4}\left(x^2 - 1\right)^{\frac{2}{3}}\right]_0^t$<br>$+ \lim_{s\to 1^+}\left[\frac{3}{4}\left(x^2 - 1\right)^{\frac{2}{3}}\right]_s^3$ | **M1** | Splits at $x = 1$ into two integrals, each written with a limit of a variable endpoint tending to 1 from the appropriate side. |
| | $= \left(0 - \frac{3}{4}\right) + (3 - 0)$ | **dM1** | Substitutes the endpoints into their antiderivative in both integrals and evaluates the limits, with the terms at the variable endpoint tending to 0. |
| | Both limits finite, so converges; $= \frac{9}{4}$ | **A1** | cso. $\frac{9}{4}$ oe with a statement that the integral converges or both limits are finite. A0 without the limit argument. |
| | | **[5]** | |
| | **Total for Question 12** | **6** | |

| Question | Scheme | Marks | Notes |
|---|---|---|---|
| **13(a)** | Integrand undefined (unbounded) at the lower limit $x = 0$ | **E1** | States the integrand is undefined, or tends to infinity, at $x = 0$. Not for a reason based on the range or the upper limit. |
| | | **[1]** | |
| **13(b)** | $\int x^{-\frac{1}{2}}(\ln x)^2\,\mathrm{d}x = 2x^{\frac{1}{2}}(\ln x)^2$ $- \int 2x^{\frac{1}{2}} \times \frac{2\ln x}{x}\,\mathrm{d}x$ | **M1 A1** | **M1** Integrates by parts with $u = (\ln x)^2$, reaching the form $px^{\frac{1}{2}}(\ln x)^2 - q\int x^{-\frac{1}{2}}\ln x\,\mathrm{d}x$. Condone coefficient slips. **A1** Correct; need not be simplified. |
| | $\int x^{-\frac{1}{2}}\ln x\,\mathrm{d}x = 2x^{\frac{1}{2}}\ln x - \int 2x^{-\frac{1}{2}}\,\mathrm{d}x$ | **dM1** | Integrates $\int x^{-\frac{1}{2}}\ln x\,\mathrm{d}x$ by parts with $u = \ln x$, reaching the form $rx^{\frac{1}{2}}\ln x - s\int x^{-\frac{1}{2}}\,\mathrm{d}x$. Condone coefficient slips. |
| | $= 2\sqrt{x}(\ln x)^2 - 4\left(2\sqrt{x}\ln x - 4\sqrt{x}\right) + c$ $= 2\sqrt{x}\left((\ln x)^2 - 4\ln x + 8\right) + c$ | **A1 AG** | Both stages correctly combined, final integral done, leading without error to the printed result, including $+c$. |
| | | **[4]** | |
| **13(c)** | $\lim_{t\to 0^+}\int_t^e x^{-\frac{1}{2}}(\ln x)^2\,\mathrm{d}x$ $= \lim_{t\to 0^+}\left[2\sqrt{x}\left((\ln x)^2 - 4\ln x + 8\right)\right]_t^e$ | **M1** | Replaces the lower limit by $t$, or another letter, and substitutes both limits into the result of (b). Not for substituting $x = 0$. |
| | $= 10\sqrt{e} - 2\sqrt{t}\left((\ln t)^2 - 4\ln t + 8\right)$ | **A1** | Correct expression in $t$, oe, e.g. expanded. |
| | As $t \to 0^+$: $\sqrt{t}(\ln t)^2 \to 0$, $\sqrt{t}\ln t \to 0$, $\sqrt{t} \to 0$ | **dM1** | Lets $t \to 0^+$, using the given result for the $\sqrt{t}(\ln t)^2$ and $\sqrt{t}\ln t$ terms and $\sqrt{t} \to 0$. |
| | Limit finite, so integral exists, $= 10\sqrt{e}$ | **A1** | cso. Exact $10\sqrt{e}$ with a statement that the limit is finite, or the integral exists. |
| | | **[4]** | |
| | **Total for Question 13** | **9** | |

| Question | Scheme | Marks | Notes |
|---|---|---|---|
| **14(a)** | $\int \frac{(\ln x)^2}{x^2}\,\mathrm{d}x = -\frac{(\ln x)^2}{x} + \int \frac{2\ln x}{x^2}\,\mathrm{d}x$ | **M1 A1** | **M1** Integrates by parts with $u = (\ln x)^2$ (or $t^2$ after $x = e^t$) to the form $p\frac{(\ln x)^2}{x} + q\int \frac{\ln x}{x^2}\,\mathrm{d}x$. Condone sign and coefficient slips.<br>**A1** Correct, need not be simplified. |
| | $\int \frac{\ln x}{x^2}\,\mathrm{d}x = -\frac{\ln x}{x} + \int \frac{1}{x^2}\,\mathrm{d}x$ | **dM1** | Integrates $\frac{\ln x}{x^2}$ (or $te^{-t}$) by parts to the form $p\frac{\ln x}{x} + q\int \frac{1}{x^2}\,\mathrm{d}x$. Condone sign and coefficient slips. |
| | $-\frac{(\ln x)^2}{x} - \frac{2\ln x}{x} - \frac{2}{x}$ | **A1** | oe; constant not needed. |
| | $\left(-\frac{(\ln a)^2}{a} - \frac{2\ln a}{a} - \frac{2}{a}\right) - (-2)$ | **dM1** | Substitutes $a$ and 1 into their antiderivative from two integrations by parts and subtracts. |
| | $2 - \frac{(\ln a)^2}{a} - \frac{2\ln a}{a} - \frac{2}{a}$ | **A1** | oe |
| | | **[6]** | |
| **14(b)** | $0$ | **B1** | |
| | | **[1]** | |
| **14(c)** | $\lim_{a\to\infty}\left(2 - \frac{(\ln a)^2}{a} - \frac{2\ln a}{a} - \frac{2}{a}\right)$<br>$= 2 - 0 - 0 - 0$ | **M1** | Lets $a \to \infty$ in their (a) of the form constant plus multiples of $\frac{(\ln a)^2}{a}$, $\frac{\ln a}{a}$, $\frac{1}{a}$, stating each such term tends to 0. |
| | $\int_1^\infty \frac{(\ln x)^2}{x^2}\,\mathrm{d}x = 2$ | **A1** | |
| | | **[2]** | |
| **Total for Question 14** | | **9** | |

| Question | Scheme | Marks | Notes |
|---|---|---|---|
| **15** | $\dfrac{5x-3}{(x^2+3)(x+1)} = \dfrac{A}{x+1} + \dfrac{Bx+C}{x^2+3}$ | **M1** | Uses a constant over $x+1$ and a linear numerator over $x^2+3$. |
| | $5x-3 = A(x^2+3)+(Bx+C)(x+1)$<br>$\Rightarrow A = \ldots,\ B = \ldots,\ C = \ldots$ | **dM1** | Multiplies through and finds all three constants by substitution or equating coefficients. Condone arithmetic slips. |
| | $A=-2,\ B=2,\ C=3$ | **A1** | |
| | $\int \left(-\dfrac{2}{x+1} + \dfrac{2x}{x^2+3}\right) \mathrm{d}x$<br>$= -2\ln(x+1) + \ln(x^2+3)$ | **M1** | Integrates their $\frac{A}{x+1}$ and $\frac{Bx}{x^2+3}$ to the forms $\alpha\ln(x+1)$ and $\beta\ln(x^2+3)$. Condone coefficient slips. |
| | $\int \dfrac{3}{x^2+3}\,\mathrm{d}x = \dfrac{3}{\sqrt{3}}\arctan\left(\dfrac{x}{\sqrt{3}}\right)$ | **M1** | Integrates their $\frac{C}{x^2+3}$ to the form $k\arctan\left(\frac{x}{\sqrt{3}}\right)$. |
| | $\ln\left(\dfrac{x^2+3}{(x+1)^2}\right) + \sqrt{3}\arctan\left(\dfrac{x}{\sqrt{3}}\right)$ | **A1** | Correct antiderivative oe, logarithms combined or not; modulus signs condoned. |
| | $\lim_{t\to\infty}\left[\ln\left(\dfrac{t^2+3}{(t+1)^2}\right) + \sqrt{3}\arctan\left(\dfrac{t}{\sqrt{3}}\right)\right]$<br>$-(\ln 3 + 0)$<br>$= 0 + \sqrt{3}\cdot\dfrac{\pi}{2} - \ln 3$ | **dM1** | Replaces $\infty$ by a limit, applies both limits, combines the logarithms into one tending to $\ln 1$, and uses $\arctan \to \frac{\pi}{2}$. Not for evaluating separate logarithms at infinity. |
| | $\frac{1}{2}\sqrt{3}\,\pi + \ln\frac{1}{3}$: $p=\frac{1}{2},\ q=\frac{1}{3}$ | **A1** | cso. Needs $q=\frac{1}{3}$ or $\ln\frac{1}{3}$; $-\ln 3$ alone scores A0. |
| | **Total for Question 15** | **8** | |

| Question | Scheme | Marks | Notes |
|---|---|---|---|
| **16(a)** | As $y \to 0^+$, $\ln y \to -\infty$, $(\ln y)^2 \to \infty$<br>Integrand unbounded at $y = 0$ | **E1** | States the integrand is unbounded or undefined at $y = 0$, referring to $\ln y$ or $(\ln y)^2$ there. |
| | | **[1]** | |
| **16(b)(i)** | $(\ln y)^2 = x^2$, $\mathrm{d}y = -e^{-x}\,\mathrm{d}x$ | **M1** | Replaces $(\ln y)^2$ by $x^2$ and $\mathrm{d}y$ by $\pm e^{-x}\,\mathrm{d}x$. Condone a sign slip in $\mathrm{d}y$. |
| | $y = 1 \Rightarrow x = 0$<br>$y \to 0^+ \Rightarrow x = -\ln y \to \infty$ | **B1** | Both limits, with $x \to \infty$ justified as $y \to 0^+$ by $-\ln y \to \infty$ or $e^{-x} \to 0$. B0 for $y = 0 \Rightarrow x = \infty$ with no limiting argument. |
| | $\int_{\infty}^{0} x^2\,(-e^{-x})\,\mathrm{d}x = \int_{0}^{\infty} x^2 e^{-x}\,\mathrm{d}x$ | **A1 AG** | Correct substituted integral with limits $\infty$ to 0 and the minus sign, reversed to the printed result without error. |
| | | **[3]** | |
| **16(b)(ii)** | $\int x^2 e^{-x}\,\mathrm{d}x = -x^2 e^{-x} + \int 2x e^{-x}\,\mathrm{d}x$ | **M1 A1** | **M1** Integrates by parts with $u = x^2$, reaching $\pm x^2 e^{-x} \pm \int kx e^{-x}\,\mathrm{d}x$. Tabular method accepted. |
| | $= -x^2 e^{-x} - 2x e^{-x} - 2e^{-x}$ | **dM1 A1** | **dM1** Integrates their $\int kx e^{-x}\,\mathrm{d}x$ by parts, reaching the form $px e^{-x} + qe^{-x}$.<br>**A1** oe; constant not needed. |
| | $\left[-e^{-x}(x^2 + 2x + 2)\right]_0^t = 2 - e^{-t}(t^2 + 2t + 2)$<br>As $t \to \infty$, $t^2 e^{-t}, te^{-t}, e^{-t} \to 0$ | **dM1** | Applies limits 0 and $t$ to their antiderivative of the form $e^{-x} \times$ quadratic, and lets $t \to \infty$ with the exponential terms tending to 0. Not for substituting $\infty$ directly. |
| | $\int_{0}^{\infty} x^2 e^{-x}\,\mathrm{d}x = 2$ | **A1** | cso. A bare answer scores nothing. |
| | | **[6]** | |
| **16(b)(iii)** | $\int_{0}^{1} (\ln y)^2\,\mathrm{d}y = 2$ | **B1ft** | ft their value from (b)(ii). |
| | | **[1]** | |
| | **Total for Question 16** | **11** | |